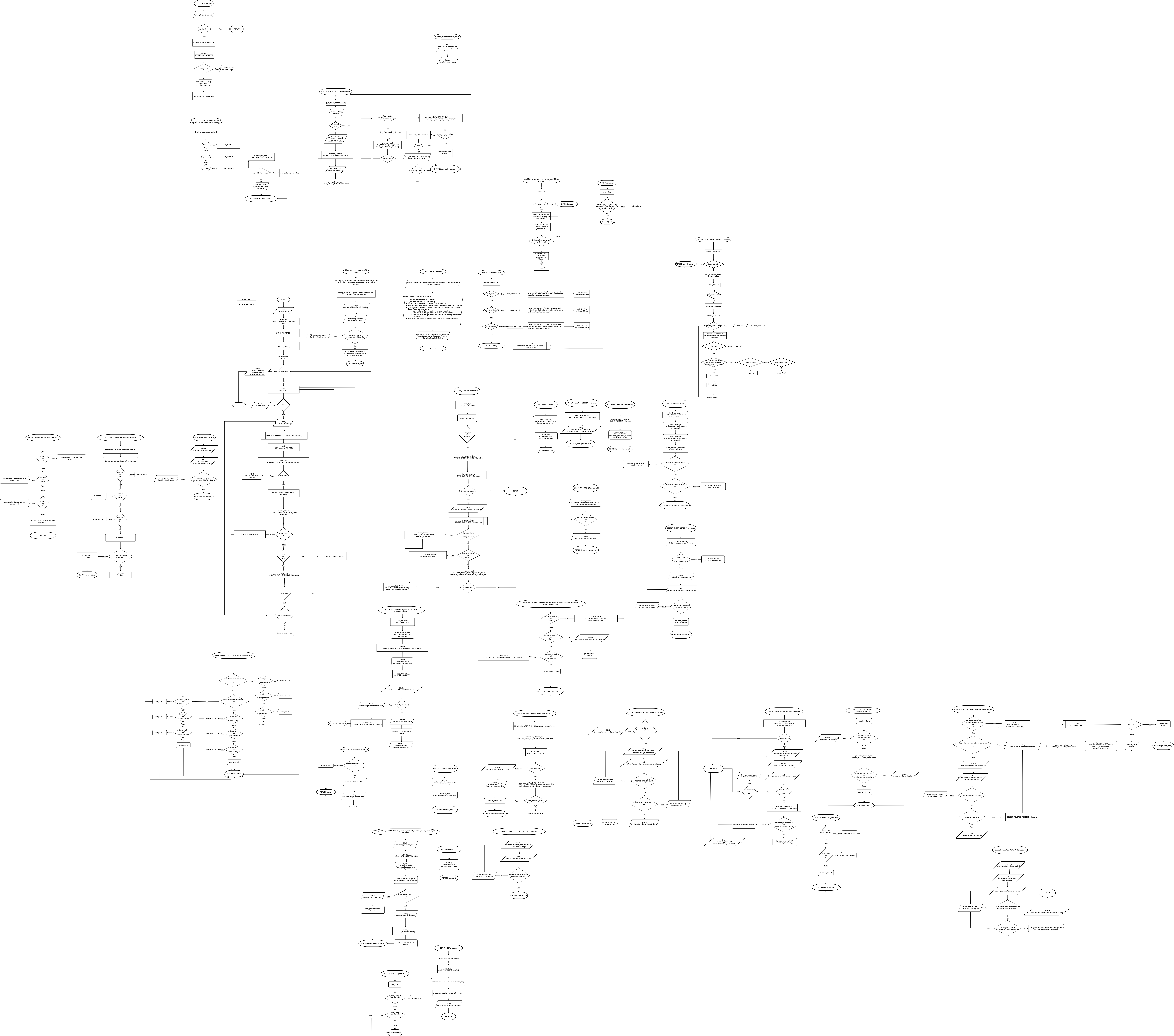
BUY_POTION(character)
Enter y to buy or n to skip
user_input = 'y'
False
RETURN
True
budget = money character has
change = budget - POTION_PRICE
change >= 0
False
You can't buy with your current budget
True
Purchase successful Your change is $(change).
money character has = change
CHECK_FOR_BADGE_CHANGE(character, actual_win_count, gym_badge_earned)
level = character's current level
level == 1
True
win_count = 2
False
level == 2
True
win_count = 3
False
level == 3
True
win_count = 4
count_left_for_badge = win_count - actual_win_count
count_left_for_badge > 0
False
gym_badge_earned = True
True
You need to win {count_left_for_badge} more time
RETURN(gym_badge_earned)
BATTLE_WITH_GYM_LEADER(character)
gym_badge_earned = False
Enter y to challenge, n to quit
character_input == 'y'
False
True
Gym leader: Welcome to the gym! Here is one rule, you can't use potions
selected_pokemon = TAKE_OUT_POKEMON(character)
You have chosen {selected_pokemon}
gym_leader_pokemon = GET_EVENT_POKEMON(character)
fight_result = FIGHT(character_pokemon, event_pokemon_info)
fight_result
False
True
attacked_result = GET_ATTACKED(event_pokemon, event_type, character_pokemon)
attacked_result
True
gym_badge_earned = CHECK_FOR_BADGE_CHANGE(character, actual_win_count, gym_badge_earned)
gym_badge_earned
False
alive = IS_ALIVE(character)
alive
False
True
Enter y if you want to proceed another battle in the gym, else n
user_input == 'y'
False
True
character's current level += 1
RETURN(gym_badge_earned)
describe_location(character_status)
Find the cell on the board that matches the character's current location
Display character's current locaion
GENERATE_STORE_LOCATIONS(board, rows, columns)
count = 0
count < 3
False
RETURN(board)
True
row = a random number between 0 (inclusive) and rows (exclusive)
column = a random number between 0 (inclusive) and columns (exclusive)
cordinate of row and column on the board
False
cordinate of row and column on the board = "Store"
True
count += 1
IS_ALIVE(character)
alive = True
At least one Pokemon in the character's Poke Ball has HP greater than 0
False
alive = False
True
RETURN(alive)
GET_CURRENT_LOCATION(board, character)
current_location = ""
board is empty
True
RETURN(current_location)
Find the maximum row and column in the board
row_index = 0
row_index < rows
Ture
Create an empty row
column_index = 0
column_index < rows
False
Print row
row_index += 1
location = coordinate of row_index and column_index on the board
location
False
row += " "
True
coordinate of row_index and column_index == character's current location
False
location == "Store"
False
location == "Gym"
True
row += "[#]"
row += "[S]"
True
row += "[G]"
current_location = location
column_index += 1
CONSTANT
POTION_PRICE = 10
START
Ask character name
character = MAKE_CHARACTER(character input)
PRINT_INSTRUCTIONS()
board = MAKE_BOARD()
achieved_goal = False
achieved_goal
True
Display "Congratulations! You have successfully finished your journey 🎉"
END
False
check = IS_ALIVE()
check
False
Display "Game Over"
True
Display current character status
DISPLAY_CURRENT_LOCATION(board, character)
direction = GET_character_CHOICE()
valid_move = VALIDATE_MOVE(board, character, direction)
valid_move
False
Display character can't go the direction
True
MOVE_CHARACTER(character, direction)
current_location = GET_CURRENT_LOCATION(board, character)
current_location == 'store'
True
BUY_POTION(character)
False
location == 'gym'
False
EVENT_OCCURRED(character)
True
battle_result = BATTLE_WITH_GYM_LEADER(character)
battle_result
False
True
character level <= 3
True
False
achieved_goal = True
MAKE_CHARACTER(character name)
character_status contains data about money, poke ball, current level, potion, current location, character name, starting pokemon
starting_pokemon = Squirtle, Charmander, Bulbasaur with their type and currentHP
Display starting pokémon list with their type
Ask which starting pokémon the character wants
character input is in the starting pokémon list
False
Tell the character about that it is not valid option
True
Put character input pokémon into poké ball with its type and HP and starting pokémon
RETURN(character_status)
PRINT_INSTRUCTIONS()
Welcome to the world of Pokemon! Embark on an exciting journey to become a Pokemon Champion.
Important notes to know before you begin:
Stores are represented by S on the map.
Gyms are represented by G on the map.
If all six of your Pokémon lose their HP, the game is over.
You can only challenge a gym leader once you have a full team of six Pokémon.
After defeating a gym leader, you will earn a badge, unlocking the next level.
Badge Requirements by Level:
Level 1: Defeat the gym leader twice to earn a badge.
Level 2: Defeat the gym leader three times to earn a badge.
Level 3: Defeat the gym leader four times to earn a badge and complete the mission.
The mission is complete when you defeat the final Gym Leader at Level 3.
Your journey will be tough, but with determination and strategy, you can become a Pokémon Champion. Good luck, Trainer!
RETURN
MAKE_BOARD(current_level)
Create an empty board
current_level == 1
True
rows, columns = 6, 6
Inside the board, mark True for the playable 4x6 rectangle and the 2 extra cells for the start and end, and mark False for all other cells
Mark "Gym" for coordinate of 5 and 5
False
current_level == 2
True
rows, columns = 8, 5
Inside the board, mark True for the playable 5x6 rectangle and the 2 extra cells for the start and end, and mark False for all other cells
Mark "Gym" for coordinate of 7 and 4
False
current_level == 3
True
rows, columns = 10, 5
Inside the board, mark True for the playable 5x8 rectangle and the 2 extra cells for the start and end, and mark False for all other cells
Mark "Gym" for coordinate of 8 and 4
False
RETURN(board)
board = GENERATE_STORE_LOCATIONS(board, rows, columns)
MOVE_CHARACTER(character, direction)
direction == up
True
current location Y-coordinate from charater -= 1
False
direction == down
True
current location Y-coordinate from charater += 1
False
direction == left
True
current location X-coordinate from charater -= 1
False
current location X-coordinate from charater += 1
RETURN
VALIDATE_MOVE(board, character, direction)
Y-coordinate = current location from character
X-coordinate = current location from character
direction == up
True
Y-coordinate -= 1
False
direction == down
True
Y-coordinate += 1
False
direction == left
True
X-coordinate -= 1
False
X-coordinate += 1
X-, Y-coordinate are in the board
False
on_the_board = False
True
on_the_board = True
RETURN(on_the_board)
GET_CHARACTER_CHOICE()
Display a numbered list of directions
Ask which direction the character wants to choose
character input is in a numbered list of directions
False
Tell the character about that it is not valid option
True
RETURN(character input)
EVENT_OCCURRED(character)
event_type = SET_EVENT_TYPE()
process_result = True
event_type == No event
True
False
event_pokemon_info = APPEAR_EVENT_POKEMON(character)
character_pokemon = TAKE_OUT_POKEMON(character)
process_result
False
RETURN
True
Display what the character's pokemon is with HP
character_choice = SELECT_EVENT_OPTION(event_type)
character_choice == change pokemon
True
character_pokemon = CHANGE_POKEMON(character, character_pokemon)
False
character_choice == use potion
True
USE_POTION(character, character_pokemon)
False
process_result = PROCEED_EVENT_OPTION(character_choice, character_pokemon, character, event_pokemon_info)
process_result
True
False
process_result = GET_ATTACKED(event_pokemon, event_type, character_pokemon)
SET_EVENT_TYPE()
event_collection = Wild pokemon, Team Rocket, Strange trainer, No event
event_type = a random event from event_collection
RETURN(event_type)
APPEAR_EVENT_POKEMON(character)
event_pokemon_info = GET_EVENT_POKEMON(character)
Display what type of event occurred and what event pokemon is with its HP
RETURN(event_pokemon_info)
GET_EVENT_POKEMON(character)
event_pokemon_collection = EVENT_POKEMON(character)
event_pokemon_info = a random pokemon from event_pokemon_collection with its type and HP
RETURN(event_pokemon_info)
EVENT_POKEMON(character)
level1_pokemon = level1 pokemon collection with their type and HP
level2_pokemon = level2 pokemon collection with their type and HP
level3_pokemon = level3 pokemon collection with their type and HP
event_pokemon_collection = level1_pokemon
current level (from character) == 2
True
event_pokemon_collection = level2_pokemon
False
current level (from character) == 3
True
event_pokemon_collection = level3_pokemon
False
RETURN(event_pokemon_collection)
TAKE_OUT_POKEMON(character)
character_pokemon = a random pokemon with its type and HP from poke ball (from character)
character_pokemon's HP == 0
True
False
Display what the character pokemon is
RETURN(character_pokemon)
SELECT_EVENT_OPTION(event_type)
character_option = Fight, Change pokemon, Use potion
event_type == Wild pokemon
True
character_option += Throw poke ball, Run
False
Display what options the character has
Ask what option the character wants to choose
character input is included in character_option
False
Tell the character about that it is not valid option
character_choice = character input
RETURN(character_choice)
PROCEED_EVENT_OPTION(character_choice, character_pokemon, character, event_pokemon_info)
character_choose == fight
True
process_result = FIGHT(character_pokemon, event_pokemon_info)
False
character_choose == Run
True
Display the character escaped from event pokemon
process_result = False
False
character_choose == Throw poke ball
True
process_result = THROW_POKE_BALL(event_pokemon_info, character)
False
process_result = False
RETURN(process_result)
GET_ATTACKED(event_pokemon, event_type, character_pokemon)
skill_collection = GET_SKILL_OF()
event_pokemon_skill = a random skill from the skill_collection
damage = MAKE_DAMAGE_STRONGER(event_type, character)
damage *= a random number from the skill damage range
skill_accuracy = GET_PROBABILITY()
Display what kind of skill the event pokemon used
skill_accuracy
False
Display the event pokemon's skill missed
True
Display the event pokemon's skill hit
character_pokemon's HP -= damage
Display how many damage character_pokemon got
process_result = CHECK_STATUS(character_pokemon)
RETURN(process_result)
MAKE_DAMAGE_STRONGER(event_type, character)
current level(from character) == 1
True
event_type == team rocket
True
stronger = 1.4
False
event_type == strange trainer
True
stronger = 1.3
False
event_type == wild pokemon
True
stronger = 1
False
stronger = 1.5
False
current level(from character) == 2
True
event_type == team rocket
True
stronger = 1.7
False
event_type == strange trainer
True
stronger = 1.5
False
event_type == wild pokemon
True
stronger = 1.3
False
stronger = 2
False
event_type == team rocket
True
stronger = 1.9
False
event_type == strange trainer
True
stronger = 1.7
False
event_type == wild pokemon
True
stronger = 1.5
False
stronger = 2.5
RETURN(stronger)
CHECK_STATUS(character_pokemon)
character pokemon's HP <= 0
False
status = True
True
character pokemon's HP = 0
Display the character pokemon fainted
status = False
RETURN(status)
GET_SKILL_OF(pokemon_type)
skills_of = skill collection depending on type with damage range
pokemon_skill = skill collection of pokemon_type
RETURN(pokemon_skill)
FIGHT(character_pokemon, event_pokemon_info)
skill_collection = GET_SKILL_OF(character_pokemon's type)
character_pokemon_skill = CHOOSE_SKILL_TO_CHALLENGE(skill_collection)
skill_accuracy = GET_PROBABILITY()
skill_accuracy
False
Display character_pokemon's skill missed
Display event pokemon's HP (from event_pokemon_info)
True
event_pokemon_status = GET_ATTACK_RESULT(character_pokemon_skill, skill_collection, event_pokemon_info, character)
event_pokemon_status
True
process_result = True
False
process_result = False
RETURN(process_result)
GET_ATTACK_RESULT(character_pokemon_skill, skill_collection, event_pokemon_info, character)
Display character_pokemon_skill hit
damage = MAKE_STRONGER(character)
damage *= a random number from the skill damage range from skill_collection
event pokemon's HP (from event_pokemon_info) -= damage
event pokemon's HP <= 0
False
Display event pokemon's HP, name
event_pokemon_status = True
True
Display event pokemon is defeated
money = GET_MONEY(character)
event_pokemon_status = False
RETURN(event_pokemon_status)
GET_PROBABILITY()
accuracy = a random result between True or False
RETURN(accuracy)
CHOOSE_SKILL_TO_CHALLENGE(skill_collection)
Display what skills character_pokemon can use with damage range
Ask what skill the character wants to use
character input is included inside character_option
False
Tell the character about that it is not valid option
True
RETURN(character input)
GET_MONEY(character)
money_range = three numbers
money = MAKE_STRONGER(character)
money *= a random number from money_range
character money(from character) += money
Display how much money the character got
RETURN
MAKE_STRONGER(character)
stronger = 1
current level (from character) == 2
True
stronger = 1.3
False
current level (from character) == 2
True
stronger = 1.3
False
RETURN(stronger)
CHANGE_POKEMON(character, character_pokemon)
Number of the character's Pokémon == 1
True
Display the character has no pokemon to switch to
False
Display the character pokemons' status from poke ball (from character)
Ask Which Pokémon the character wants to switch to
character input is included in the character pokemon list
False
Tell the character about that it is not valid option
True
character input pokemon' HP == 0
True
Tell the character about the pokemon has 0 HP
False
Display the character pokemon is switching to
character_pokemon = character input
RETURN(character_pokemon)
USE_POTION(character, character_pokemon)
validate_potion = CHECK_POTION(character, character_pokemon)
validate_potion
False
True
Display how many potions the character has (from character)
Display character_pokemon's status
Ask the character wants to use a potion
character input == yes
False
character input == no
True
False
Tell the character about that it is not valid option
True
pokemon_maximum_hp = LEVEL_MAXIMUM_HP(character)
character_pokemon's HP >= pokemon_maximum_hp - 5
False
character_pokemon's HP += 5
True
character_pokemon's HP = pokemon_maximum_hp
Display how many potions left and what character_pokemon's HP
RETURN
CHECK_POTION(character, character_pokemon)
validation = False
the amount of potion the character has == 0
True
Display the character has no potion
False
pokemon_maximum_hp = LEVEL_MAXIMUM_HP(character)
character_pokemon's HP == pokemon_maximum_hp
True
Display character_pokemon has full HP
False
validation = True
RETURN(validation)
LEVEL_MAXIMUM_HP(character)
current level (from character) == 1
True
maximum_hp = 30
False
current level == 2
True
maximum_hp = 50
False
maximum_hp = 80
RETURN(maximum_hp)
THROW_POKE_BALL(event_pokemon_info, character)
event pokemon's HP (from event_pokemon_info) <= 5
False
Display the character failed to catch the event pokemon
run_or_not = GET_PROBABILITY()
run_or_not
True
process_result = True
False
True
Total pokemon number the character has == 6
False
Display what pokemon the character caught
pokemon_maximum_hp = LEVEL_MAXIMUM_HP(character)
Add the event pokemon to the character pokemon collection with its type and a half of pokemon_maximum_hp
process_result = False
RETURN(process_result)
True
Display the character has up to 6 pokemons
Ask the character wants to release one character pokemon
character input is yes or no
False
Tell the character about that it is not valid option
True
character input is no
False
SELECT_RELEASE_POKEMON(character)
True
Tell the event pokemon broke free
SELECT_RELEASE_POKEMON(character)
Display all of character's pokemons with HP
Tell the character can't choose starting pokemon
Ask what pokemon the character release
The character input is included in the character's Pokémon collection
False
Tell the character about that it is not valid option
True
The character input is the character's starting pokemon
True
False
Remove the character input pokemon's information from the character pokemon collection
Display the character released character input pokemon
RETURN